

ECON 340: Economics of the Family

TA Session 9

Vaidehi Parameswaran (Northwestern Economics)

November 2025

- ▶ The Gender Wage Gap and Domestic Violence
- ▶ Intimate Partner Violence: The Influence of Job Opportunities for Men and Women
- ▶ Gender Violence, Enforcement, and Human Capital: Evidence from Women's Justice Centers in Peru

Aizer (2010). *The Gender Wage Gap and Domestic Violence. American Economic Review*

- ▶ **Question:** Does narrowing the male–female wage gap reduce intimate partner violence (IPV)?
- ▶ **Key challenges:**
 - ▷ Reverse causality
 - ▷ Omitted variable bias

- ▶ Three-quarters of all violence against women is perpetrated by domestic partners.
- ▶ Disproportionate incidence among poor women.
- ▶ \$5.8 billion spent annually in the United States on medical care and productivity losses.

Household Bargaining

- ▶ Increases in women's relative wages $\implies$ better outside options.
- ▶ Improved outside options for women $\implies$ higher bargaining power $\implies$ lower IPV.
- ▶ Alternative theories to test against:
 - ▷ **Male backlash:** higher female earnings provoke more IPV.
 - ▷ **Exposure reduction:** more female work hours reduce time with partner, mechanically lowering IPV.

Identification

Two main challenges:

- ▶ Lack of objective measures of domestic violence collected consistently over time.

Identification

Two main challenges:

- ▶ Lack of objective measures of domestic violence collected consistently over time.
Solution: Construct a new measure of female hospitalizations for assault in California from 1990–2003 (not self-reported).
- ▶ Difficulty constructing measures of *relative* labor market conditions that do not reflect the underlying characteristics of male and female workers (which could themselves be functions of violence or correlated unobservables such as education).
Solution: Use the historical sex and race segregation of industries to construct county-level women's wages based on *national* wage changes in industries dominated by women.



$$\overline{w_{grcy}} = \sum_j \gamma_{grcj} w_{-cyj}$$

Regression Form & Threats

$$DV_{cy} = \alpha_c + \lambda_y + \beta_1 \text{WageRatio}_{cy} + \beta_2 DV_{c,y-1} + \mathbf{X}_{cy}^\top \delta + \varepsilon_{cy}. \quad (1)$$

- ▶ **Key regressor:** local male–female **wage gap** (WageRatio); constructed using gender-specific Bartik-style demand shocks.
- ▶ **Controls:** unemployment rate, violent **crime** rate, per-capita income, race shares, population; plus **county** and **year** fixed effects.
- ▶ **Exogeneity:** national industry wage growth (used in the Bartik) is plausibly exogenous to local IPV shocks, conditional on fixed effects and controls.

Results

TABLE 2—IMPACT OF WAGES ON DOMESTIC VIOLENCE—MAIN SPECIFICATION

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)	ln(male assaults) (5)	ln(female assaults) (6)
<i>Panel A. Ratio of wages</i>						
Female/male wage	-1.469 [0.673]	-0.831 [0.313]	-0.813 [0.317]	-0.023 [0.072]	-0.257 [0.284]	0.119 [0.562]
Female/male wage × weekend						-1.15 [0.444]
Observations	984	982	982	887	982	616
R ²	0.91	0.95	0.96	0.99	0.99	0.96
<i>Panel B. Linear difference in wages</i>						
Male wage–female wage	0.0047 [0.0020]	0.0024 [0.0009]	0.0024 [0.0009]	0 [0.000]	0.0009 [0.0008]	-0.0003 [0.0017]
(Male–female wage) × weekend						0.0031 [0.0015]
Observations	984	982	982	887	982	616
R ²	0.91	0.95	0.96	0.99	0.99	0.96
County, year, race fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County and race specific linear time trends	Yes	Yes	Yes	Yes	Yes	Yes
ln(population), ln(nonintimate homicides)		Yes	Yes	Yes	Yes	Yes
Unemployment rate and ln(per capita income)		Yes	Yes	Yes	Yes	Yes

Results

TABLE 2—IMPACT OF WAGES ON DOMESTIC VIOLENCE—MAIN SPECIFICATION

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)	ln(male assaults) (5)	ln(female assaults) (6)
<i>Panel A. Ratio of wages</i>						
Female/male wage	−1.469 [0.673]	−0.831 [0.313]	−0.813 [0.317]	−0.023 [0.072]	−0.257 [0.284]	0.119 [0.562]
Female/male wage × weekend						−1.15 [0.444]
Observations	984	982	982	887	982	616
R ²	0.91	0.95	0.96	0.99	0.99	0.96
<i>Panel B. Linear difference in wages</i>						
Male wage−female wage	0.0047 [0.0020]	0.0024 [0.0009]	0.0024 [0.0009]	0 [0.000]	0.0009 [0.0008]	−0.0003 [0.0017]
(Male−female wage) × weekend						0.0031 [0.0015]
Observations	984	982	982	887	982	616
R ²	0.91	0.95	0.96	0.99	0.99	0.96
County, year, race fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County and race specific linear time trends	Yes	Yes	Yes	Yes	Yes	Yes
ln(population), ln(nonintimate homicides)		Yes	Yes	Yes	Yes	Yes
Unemployment rate and ln(per capita income)		Yes	Yes	Yes	Yes	Yes

Results

There seems to be no male backlash.

TABLE 2—IMPACT OF WAGES ON DOMESTIC VIOLENCE—MAIN SPECIFICATION

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)	ln(male assaults) (5)	ln(female assaults) (6)
<i>Panel A. Ratio of wages</i>						
Female/male wage	−1.469 [0.673]	−0.831 [0.313]	−0.813 [0.317]	−0.023 [0.072]	−0.257 [0.284]	0.119 [0.562]
Female/male wage × weekend						−1.15 [0.444]
Observations	984	982	982	887	982	616
R^2	0.91	0.95	0.96	0.99	0.99	0.96
<i>Panel B. Linear difference in wages</i>						
Male wage−female wage	0.0047 [0.0020]	0.0024 [0.0009]	0.0024 [0.0009]	0 [0.000]	0.0009 [0.0008]	−0.0003 [0.0017]
(Male−female wage) × weekend						0.0031 [0.0015]
Observations	984	982	982	887	982	616
R^2	0.91	0.95	0.96	0.99	0.99	0.96
County, year, race fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County and race specific linear time trends	Yes	Yes	Yes	Yes	Yes	Yes
ln(population), ln(nonintimate relationships)		Yes	Yes	Yes	Yes	Yes

Results

It's not about exposure reduction.

TABLE 2—IMPACT OF WAGES ON DOMESTIC VIOLENCE—MAIN SPECIFICATION

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)	ln(male assaults) (5)	ln(female assaults) (6)
<i>Panel A. Ratio of wages</i>						
Female/male wage	−1.469 [0.673]	−0.831 [0.313]	−0.813 [0.317]	−0.023 [0.072]	−0.257 [0.284]	0.119 [0.562]
Female/male wage × weekend						−1.15 [0.444]
Observations	984	982	982	887	982	616
R^2	0.91	0.95	0.96	0.99	0.99	0.96
<i>Panel B. Linear difference in wages</i>						
Male wage−female wage	0.0047 [0.0020]	0.0024 [0.0009]	0.0024 [0.0009]	0 [0.000]	0.0009 [0.0008]	−0.0003 [0.0017]
(Male−female wage) × weekend						0.0031 [0.0015]
Observations	984	982	982	887	982	616
R^2	0.91	0.95	0.96	0.99	0.99	0.96
County, year, race fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
County and race specific linear time trends	Yes	Yes	Yes	Yes	Yes	Yes

Test that Relative Wages Matter

Relative wages (outside options) predict IPV; not simply absolute wages.

TABLE 4—IMPACT OF RELATIVE WAGES ON DOMESTIC VIOLENCE: ALTERNATIVE SPECIFICATIONS

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)
ln (female wage)	−0.781 [0.559]			
ln (male wage)	0.956 [0.516]			
Female/male wage		−0.697 [0.351]	−0.964 [0.355]	0.019 [0.196]
Observations	982	955	804	776
R^2	0.96	0.96	0.96	0.99
<i>Test that female and male wages are equal and opposite in value</i>				
$F(1, 37)$	0.06			
p -value	0.81			

Notes: Robust standard errors clustered on county in brackets. Column 1 is based on an OLS fixed effect regression; in column 2, I instrument for the wage ratio using statewide growth in employment by industry weighted by the county-specific shares in these industries; in columns 3 and 4 the wage ratio is derived from changes in the industrial composition of the county over time; in column 4 are results of a falsification exercise.

What Do We Learn?

- ▶ Narrowing the wage gap may have effects beyond equity and fairness—tangible health benefits.
- ▶ Outside options likely matter.
- ▶ Policy levers that raise women's human capital, access to high-wage sectors, and pay equity.
- ▶ Intergenerational effects of these policies might be large.

- ▶ The Gender Wage Gap and Domestic Violence
- ▶ Intimate Partner Violence: The Influence of Job Opportunities for Men and Women
- ▶ Gender Violence, Enforcement, and Human Capital: Evidence from Women's Justice Centers in Peru

A Global Perspective

- ▶ IPV is pervasive around the world.
- ▶ Increasingly seen as a necessary component of the development agenda.
- ▶ Generally higher in low-income countries—about 65.6% in Central Africa and 42% in South Asia.
- ▶ Serious effects on the physical and mental health of women, and on children.

Bhalotra, Kambhampati, Rawlings & Siddique (2021). *Intimate Partner Violence: The Influence of Job Opportunities for Men and Women, World Bank Economic Review*

- ▶ How does IPV vary with macroeconomic conditions?
- ▶ In particular, male and female unemployment rates.
- ▶ Compare data for 31 developing countries.

Why This Matters

- ▶ Consider shocks to male and female unemployment.
- ▶ Omitting either of these will bias estimates. Why?

Why This Matters

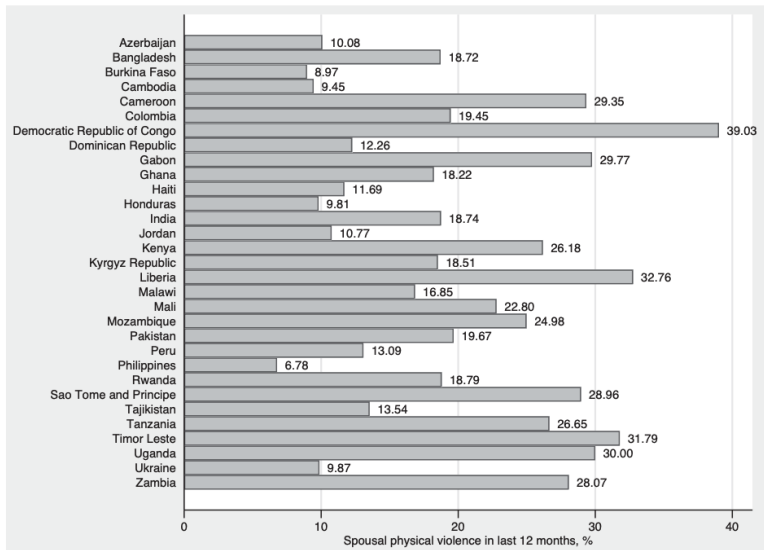
- ▶ Consider shocks to male and female unemployment.
- ▶ Omitting either of these will bias estimates. Why?
- ▶ These shocks are likely correlated.
- ▶ Important for policy: many interventions in developing countries aim to increase women's employment.
- ▶ Ex ante unclear—male backlash may arise.

- ▶ Individual data from the Demographic and Health Surveys (DHS).
- ▶ Standardized surveys that are comparable across countries.
- ▶ Survey design aims to minimize underreporting of IPV; reports are not observable to husbands.
- ▶ Years 2005–2016, 31 countries.

Unemployment Shocks

- ▶ Use time-series methods to extract the cyclical component of unemployment rates.
- ▶ Female and male shocks to unemployment are positively correlated.
- ▶ But shocks to female unemployment are larger in magnitude.

Incidence of Physical Violence



(a) Incidence of physical violence across countries

Regression Specification

- ▶ Equation:

$$\text{IPV}_{ijt} = \alpha + \beta_1 U_{j,t-1}^F + \beta_2 U_{j,t-1}^M + \mathbf{X}_{ict}^\top \delta + \gamma_j + \lambda_t + \varepsilon_{ijt} \quad (2)$$

- ▶ Variables of interest: detrended unemployment shocks U^M, U^F lagged one period.
- ▶ Controls include education, urban location, age, religion, etc.
- ▶ Standard errors clustered at the country level.
- ▶ Note: This is panel OLS (associational), not a causal identification strategy.

Regression Results

Holding the female unemployment rate constant, a 1% rise in male unemployment is associated with a 0.5 p.p. increase in IPV.

Table 1. Impact of Cyclical Fluctuations in Male and Female Unemployment Rates on IPV

	Main sample				Restricted sample: Religion controls	
	(1)	(2)	(3)	(4)	(5)	(6)
A: Physical violence						
$\ln(U^f)$	0.171 (0.148)	0.347* (0.178)	-0.534*** (0.081)	-0.519*** (0.083)	-0.502*** (0.107)	-0.506*** (0.108)
$\ln(U^m)$	-0.312** (0.151)	-0.479** (0.182)	0.508*** (0.086)	0.497*** (0.088)	0.458*** (0.163)	0.462*** (0.164)
Observations	381,730	381,730	381,730	348,165	206,641	206,641
P-value F-test	0.107	0.026	0.000	0.000	0.002	0.002
Mean violence	0.182	0.182	0.182	0.181	0.187	0.187
Mean $\ln(U^f)$	0.018	0.018	0.018	0.024	0.062	0.062
Mean $\ln(U^m)$	0.008	0.008	0.008	0.012	0.019	0.019
B: Sexual violence						
$\ln(U^f)$	0.038 (0.126)	0.089 (0.142)	-0.172 (0.107)	-0.172 (0.109)	-0.159 (0.116)	-0.160 (0.115)
$\ln(U^m)$	-0.150 (0.112)	-0.186 (0.135)	0.141 (0.107)	0.140 (0.107)	0.105 (0.159)	0.107 (0.159)
Observations	275,718	275,718	275,718	268,173	199,929	199,929
P-value F-test	0.436	0.325	0.150	0.154	0.336	0.332
Mean violence	0.068	0.068	0.068	0.067	0.075	0.075
Mean $\ln(U^f)$	0.021	0.021	0.021	0.022	0.061	0.061

Regression Results

Holding the male unemployment rate constant, a 1% rise in female unemployment is associated with a 0.52 p.p. *decrease* in IPV.

Table 1. Impact of Cyclical Fluctuations in Male and Female Unemployment Rates on IPV

	Main sample				Restricted sample: Religion controls	
	(1)	(2)	(3)	(4)	(5)	(6)
A: Physical violence						
$\ln(U^f)$	0.171 (0.148)	0.347* (0.178)	-0.534*** (0.081)	-0.519*** (0.083)	-0.502*** (0.107)	-0.506*** (0.108)
$\ln(U^m)$	-0.312** (0.151)	-0.479** (0.182)	0.508*** (0.086)	0.497*** (0.088)	0.458*** (0.163)	0.462*** (0.164)
Observations	381,730	381,730	381,730	348,165	206,641	206,641
P-value F-test	0.107	0.026	0.000	0.000	0.002	0.002
Mean violence	0.182	0.182	0.182	0.181	0.187	0.187
Mean $\ln(U^f)$	0.018	0.018	0.018	0.024	0.062	0.062
Mean $\ln(U^m)$	0.008	0.008	0.008	0.012	0.019	0.019
B: Sexual violence						
$\ln(U^f)$	0.038 (0.126)	0.089 (0.142)	-0.172 (0.107)	-0.172 (0.109)	-0.159 (0.116)	-0.160 (0.115)
$\ln(U^m)$	-0.150 (0.112)	-0.186 (0.135)	0.141 (0.107)	0.140 (0.107)	0.105 (0.159)	0.107 (0.159)
Observations	275,718	275,718	275,718	268,173	199,929	199,929
P-value F-test	0.436	0.325	0.150	0.154	0.336	0.332
Mean violence	0.068	0.068	0.068	0.067	0.075	0.075
Mean $\ln(U^f)$	0.021	0.021	0.021	0.022	0.061	0.061

What Could Be Happening Here?

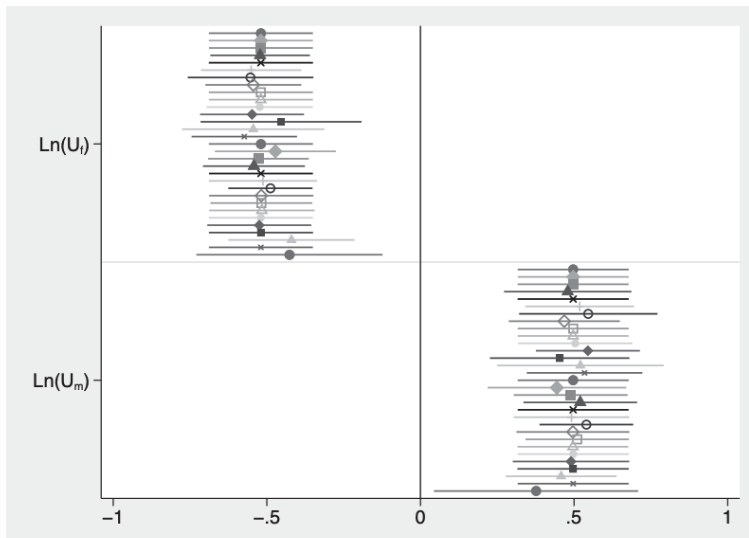
- ▶ Men react adversely to improvements in women's employment prospects, conditional on their own.
- ▶ Consistent with male backlash.
- ▶ When traditional gender roles are threatened, some men resort to violence to re-establish authority.
- ▶ This has been documented in studies beyond IPV.

Heterogeneity by Access to Divorce

Striking separation by divorce rights—effects are driven by countries where divorce is difficult for women.

Table 2. Heterogeneity by Access to Divorce

	Physical violence (I)	Sexual violence (II)	Any violence (III)
A: Equal access to divorce			
$\ln(U_f)$	1.433*** (0.087)	1.526*** (0.019)	1.211*** (0.051)
$\ln(U_m)$	−0.789*** (0.045)	−0.815*** (0.009)	−0.662*** (0.023)
Observations	95,040	35,794	35,736
X controls	Y	Y	Y
Country and year FE	Y	Y	Y
B: Women lack equal access to divorce			
$\ln(U_f)$	−0.517*** (0.107)	−0.290** (0.131)	−0.631** (0.243)
$\ln(U_m)$	0.562*** (0.077)	0.200 (0.121)	0.597*** (0.191)
Observations	253,125	232,379	230,781
X controls	Y	Y	Y
Country and year FE	Y	Y	Y



(a) Variation in estimated coefficients across sample – physical violence

Discussion (I)

- ▶ The paper finds a systematic association between unemployment rates and IPV.
- ▶ An opposing pattern: IPV appears higher when men are unemployed and when women are employed (*ceteris paribus*).
- ▶ This contrasts with Aizer (2010) (and much of the US/UK literature)—why?

Discussion (I)

- ▶ The paper finds a systematic association between unemployment rates and IPV.
- ▶ An opposing pattern: IPV appears higher when men are unemployed and when women are employed (*ceteris paribus*).
- ▶ This contrasts with Aizer (2010) (and much of the US/UK literature)—why?
- ▶ The US and the UK are settings where male-breadwinner norms have weakened over time.
- ▶ Women are also better able to exit relationships.

Discussion (II)

- ▶ In many developing countries, patriarchal norms remain strong.
- ▶ Inequality in access to divorce persists and women's labor-force participation remains low.
- ▶ Psychic costs of male unemployment and of women's employment may be high.
- ▶ Exposure-reduction stories are less credible here given the opposing effects on female unemployment.

Implications for Bargaining Models

- ▶ Despite not being causal evidence, the paper documents these conditional correlations clearly.
- ▶ Standard bargaining models emphasize that the intra-household balance of power is highly sensitive to outside options.
- ▶ But these models have different implications where outside options cannot be exercised easily because of legal or social constraints (explicit or implicit).
- ▶ Unfortunately, in many developing countries, women may lack the financial means or social support to end the partnership.
- ▶ IPV is recurrent—some women do not leave even after being subjected to violence the first time.
- ▶ Context matters for policy; effects need not be unambiguous.

- ▶ The Gender Wage Gap and Domestic Violence
- ▶ Intimate Partner Violence: The Influence of Job Opportunities for Men and Women
- ▶ Gender Violence, Enforcement, and Human Capital: Evidence from Women's Justice Centers in Peru

What Else Might Affect IPV in Developing Countries?

- ▶ The justice system in many developing countries is weak.
- ▶ Women cannot rely on it to protect them from IPV:
 - ▷ Very high rates of under-reporting of gender-based violence
 - ▷ Low rates of arrests for crimes against women
- ▶ Lack of trust in formal institutions hinders reporting and enforcement.

Sviatschi & Trako (2024). *Gender violence, enforcement, and human capital: Evidence from women's justice centers in Peru.* *Journal of Development Economics*, 168, 103262.

Question 1: Does improving law-enforcement responses to gender-based violence increase reporting and deter future violence?

Question 2: Are there intergenerational effects of improving women's access to justice on their children's human capital?

Context and Setting

- ▶ Peru: IPV is particularly acute—among the highest rates in Latin America and the Caribbean.
- ▶ 1994 Inter-American Convention on the Prevention, Punishment, and Eradication of Violence against Women.
- ▶ Peru altered the jurisdiction of its police and justice system to encompass domestic and sexual violence complaints and resolution.
- ▶ In 1999, this led to the creation of Women's Justice Centers (WJCs).

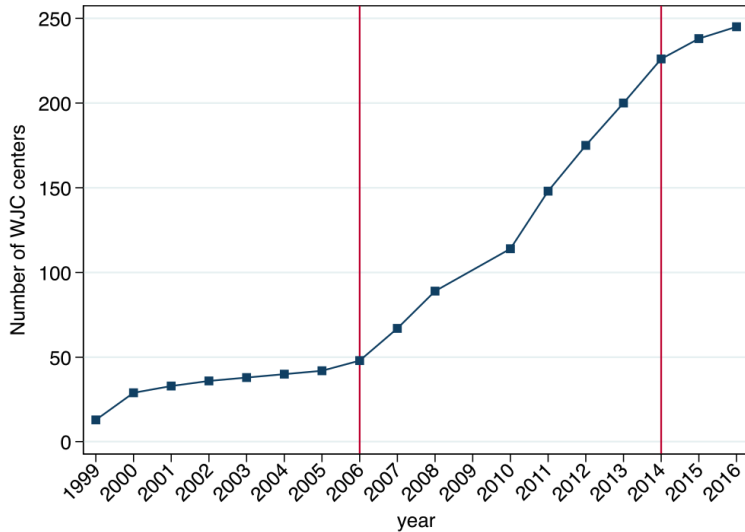
Women's Justice Centers (WJCs)

- ▶ Free-of-charge specialized public centers.
- ▶ Objectives: strengthen the justice system's capacity to detect and assist victims of gender-based violence.
- ▶ Assistance through access to legal, social, and psychological resources.
- ▶ Reduces the time and hassle cost of filing complaints.
- ▶ Personnel are trained to handle GBV cases and to reduce potential mistreatment of victims.

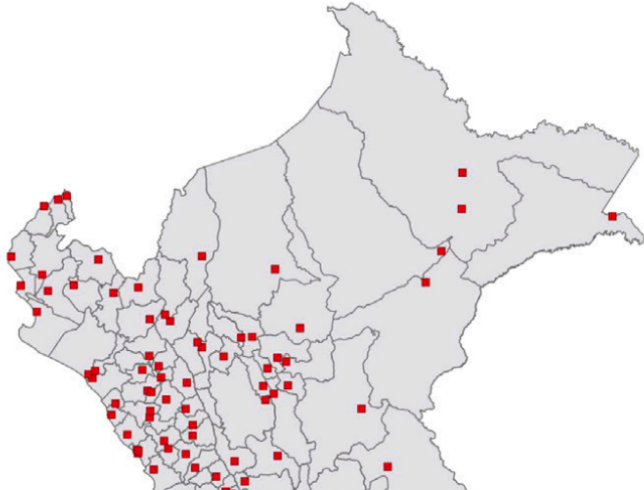
Steps at a WJC

- ▶ An admissions officer identifies the reason for the visit and creates an administrative record.
- ▶ Trained psychologists interview the victim to assess psychological state and provide emotional support; referrals to specialized health centers when necessary.
- ▶ The legal team offers assistance, participates in the investigation, and supports prosecution.
- ▶ Social workers provide counseling and help design a safety plan for victims.

Growth of WJCs in Peru



d. WJC centers in 2014



- ▶ DHS surveys (as in Bhalotra et al. 2021).
- ▶ School-level datasets: enrollment by grade and gender; test scores.
- ▶ Directory of WJCs: location and opening dates.
- ▶ Hospital data from the Peruvian Ministry of Health—deaths from aggression and female hospitalizations.
- ▶ Construct distance to the nearest WJC for each DHS cluster; define exposure within 1 km.

Identification Challenge

- ▶ Key challenge: non-random placement of WJCs.
 - ▷ **Approach 1:** Descriptive evidence that placement was driven by population density rather than preexisting violence.
 - ▷ **Approach 2:** Use data to test for endogeneity directly.

District Size Appears to Drive Placement

Table 1

Placement of WJC centers in the district.

Source: MIMP (*Ministerio de la Mujer y Poblaciones Vulnerables*).

Dependent variables	WJC in district, by 2014		Added WJC in district during 2006–2014			
	(1)	(2)	(3)	(4)	(5)	(6)
# Police Stations	0.0093 (0.0123)	0.0064 (0.0125)	−0.0098 (0.0130)	−0.0105 (0.0130)	−0.0446** (0.0205)	−0.0128 (0.0132)
# of Health Establishments	0.0024 (0.0016)	0.0024 (0.0016)	0.0001 (0.0013)	0.0000 (0.0013)	0.0005 (0.0012)	0.0001 (0.0013)
# Criminal Attorney Offices	0.0267 (0.0289)	0.0250 (0.0284)	0.0030 (0.0251)	0.0027 (0.0249)	−0.0100 (0.0206)	0.0016 (0.0246)
# Family Attorney Offices	0.0122 (0.0518)	0.0128 (0.0512)	0.0160 (0.0488)	0.0162 (0.0485)	0.0069 (0.0431)	0.0168 (0.0482)
# Courts	0.0236 (0.0145)	0.0235 (0.0144)	0.0147 (0.0135)	0.0145 (0.0135)	0.0122 (0.0110)	0.0144 (0.0134)
Log. Population, 2000	0.0744*** (0.0112)	0.0723*** (0.0119)	0.0740*** (0.0102)	0.0717*** (0.0101)	0.1167*** (0.0189)	0.0714*** (0.0111)
△ Primary Enrollment, (1998–2005)			0.0001 (0.0003)		0.0003 (0.0004)	0.0002 (0.0003)
△ Secondary Enrollment, (1998–2005)				−0.0001 (0.0001)	−0.0002 (0.0002)	−0.0001 (0.0001)
Domestic Violence, 2000					0.1065 (0.0839)	
CCT <i>Juntos</i> in the district		−0.0605** (0.0242)				−0.0451* (0.0249)
# Households with CCT <i>Juntos</i> , 2014		0.0000 (0.0000)				0.0000 (0.0000)
Observations	1,843	1,838	1,843	1,843	700	1,838
R-squared	0.3671	0.3708	0.1635	0.1638	0.1555	0.1670
Department FE	YES	YES	YES	YES	YES	YES

Empirical Strategy

- ▶ Difference-in-Differences.
- ▶ Exploit variation from differential timing in the opening of WJCs and spatial variation in exposure to a WJC.
- ▶ Individual-level regression:

$$Y_{idt} = \alpha + \beta \text{WJC}_{idt} + \mathbf{X}_{idt}^{\top} \delta + \gamma_d + \lambda_{pt} + \varepsilon_{idt} \quad (3)$$

- ▶ Key identifying assumption: in the absence of WJCs, treated and control households within the same province would have followed similar trends.
- ▶ District-level regression:

$$Y_{dt} = \alpha + \beta \text{WJC}_{dt} + \mathbf{X}_{dt}^{\top} \delta + \gamma_d + \lambda_{pt} + \varepsilon_{dt} \quad (4)$$

- ▶ FE logic: unobserved time-varying shocks within a province are uncorrelated with the within-province rollout of WJCs after conditioning on fixed effects and controls.

WJCs Reduce Self-Reported IPV

Table 2

The effect of WJC centers on self-reported domestic violence (2006–2014).

Source: Peru DHS 2006–2014.

Dep. variable	Self-reported domestic violence		
Sample	All women	All women	Only urban clusters
Controls	Standard (1)	District trends (2)	Standard (3)
WJC within 1km	−0.022** (0.010)	−0.018* (0.011)	−0.029*** (0.010)
Observations	64,363	64,363	38,395
Number of districts	1,167	1,167	485
Mean dep. var	0.390	0.390	0.399
District FE	YES	YES	YES
Province*Year FE	YES	YES	YES
Covariates	YES	YES	YES

WJCs Reduce Deaths and Hospitalizations from Violence

Table 3

WJC centers and gender-based violence at the district level.

Dep. var.	Log(# female deaths due to aggression) 2007, 2012–2014			Log(# female mental health problems) 2006–2016		
	(1)	(2)	(3)	(4)	(5)	(6)
WJC in district	–0.074** (0.031)	–0.075** (0.031)	–0.078** (0.031)	–0.0781* (0.043)	–0.0875** (0.043)	–0.101** (0.045)
Observations	7,380	7,368	7,368	20,306	20,262	20,262
Number of districts	1,846	1,842	1,842	1,846	1,842	1,842
Mean dep. var.	0.080	0.080	0.080	5.25	5.25	5.25
District FE	YES	YES	YES	YES	YES	YES
Province-Year FE	YES	YES	YES	YES	YES	YES
Population		YES	YES		YES	YES
District trends			YES			YES

WJCs Increase School Attendance

Table 4

The effect of WJC centers on children's primary school attendance (2006–2014).

Source: Peru DHS 2006–2014.

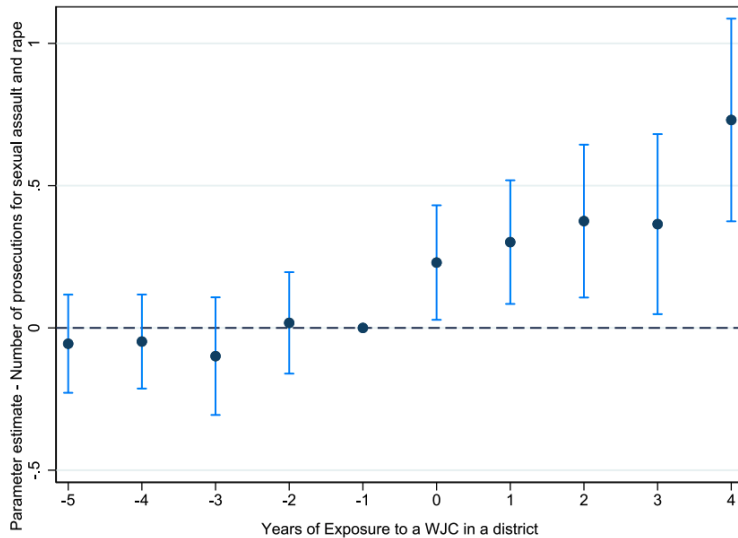
Dep. variable	Currently attending primary level		
	All children 6-11 y.o	All children 6-11 y.o	Only urban clusters
Controls	Standard (1)	District trends (2)	Standard (3)
WJC within 1km	0.019** (0.008)	0.018* (0.009)	0.027*** (0.009)
Observations	48,703	48,703	25,391
Number of districts	1,159	1,159	485
Mean dep. var	0.970	0.970	0.971
District FE	YES	YES	YES
Province*Year FE	YES	YES	YES
Covariates	YES	YES	YES

2 p.p. increase in attendance for children exposed to WJCs.

Potential Mechanisms

- ▶ Exposure to household violence: split by whether the grandmother was exposed to violence—it matters.
- ▶ Intra-household bargaining: suggestive evidence that women near WJCs are more likely to make joint decisions.
- ▶ Incapacitation: compare households where the father is present vs. absent; no meaningful differences.
- ▶ Perceptions of safety in the area: no reduction in other crimes.

Persistent Positive Effects on Prosecutions



- ▶ The opening of a WJC has positive effects on women's and children's outcomes.
- ▶ Effects are concentrated in very local areas (within 1 km).
- ▶ WJCs are becoming popular around the world.
- ▶ They encourage reporting and increase the probability of prosecution.
- ▶ Policy lesson: law and enforcement matter.
- ▶ Consider the *spillover* effects on children when evaluating policies.